YIFEI SUN

Principal Software Engineer · AI & Developer Platform Architecture · 13 Years

(617) 922-4823
sun1fei@gmail.com
linkedin.com/in/yifei-sun-a789861b

SUMMARY

Staff/Principal-level platform engineer with 13 years of experience building developer platforms, API extensibility layers, and AI-integrated systems. Currently leading MCP server development at CarGurus, exposing platform capabilities into ChatGPT via the OpenAI Apps SDK — designing the server platform, API surface, and production rollout strategy. Built and launched a federated GraphQL gateway (Netflix DGS) that unified data access for frontend, backend, and AI consumers. Founding backend engineer of an innovation lab, with a track record of shipping 0→1 products and leading cross-org platform initiatives spanning 10+ teams.

LEADERSHIP HIGHLIGHTS

- Built MCP server platform exposing product capabilities into ChatGPT — concept to production as sole backend engineer
- Led cross-org platform initiatives spanning 10+ teams; defined API design and extensibility standards
- Designed GraphQL federation gateway unifying data for frontend, backend, and AI consumers from a single graph
- Mentored senior engineers; shaped platform strategy, roadmaps, and engineering best practices across org

Technologies: Java, Spring Boot, Python, TypeScript/Node.js, GraphQL (Netflix DGS) · AWS, Kubernetes, Kafka, Redis, PostgreSQL · MCP/LLM Integration, API Platforms, Developer Extensibility, Event-Driven, Microservices · Datadog, Grafana · CI/CD, Docker

INDEPENDENT PROJECT

GenAI Agent Platform & Evaluation System

- Architected an agent platform with hybrid rules + multi-path LLM inference (fast vs. deep), tool-calling orchestration, and a policy overlay — producing explainable, grounded recommendations over heterogeneous data sources.
- Built an agent evaluation harness ("Prompt Lab") for end-to-end simulation, offline backtesting, and prompt/parameter search (grid & Bayesian optimization) with observability, multi-provider LLM support, and model quality regression detection.
- Designed a configurable framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop — enabling safe, iterative agent improvement with full traceability.

EXPERIENCE

CarGurus — Principal Software Engineer / Founding Backend Engineer, Innovation Lab Oct 2019 – Present

ChatGPT Apps SDK — MCP Server Platform (0→1)

- Sole backend engineer who designed and built an MCP server platform compatible with the OpenAI Apps SDK, exposing CarGurus' product capabilities into ChatGPT — from concept to working demo in days, iterating rapidly with PMs and directors.
- Defined the API surface, tool schemas, authentication model, and production rollout strategy; designed for safe, incremental adoption alongside existing REST and GraphQL APIs and clients.
- Became the key engineering representative for AI integration strategy, shaping the company's technical roadmap for LLM-powered product features across multiple teams.

GraphQL Federation Gateway — Unified API for AI & Product Consumers

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, creating a unified API layer that serves frontend, backend, and AI consumers (including the new AI chatbot/shopping assistant) from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load tests and staging integration tests. Migrating legacy REST endpoints to support the expanding AI platform.

High-Traffic Legacy API Modernization

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm); consolidated 12 downstream integrations into 5 well-defined services.

  p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

Digital Shopping & Online Leads Products (0→1, Multiple Launches)

- As founding backend engineer of the innovation lab, shipped multiple 0→1 products from concept to production — including a digital commerce platform built during the pandemic using microservice and event-driven architecture.
- Enabled the company to rapidly test, launch, and safely retire product experiments without destabilizing core systems.

Cross-Org Platform & API Leadership

- Established patterns for API design, extensibility, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and decision records driving consensus across platform and product teams.
- Mentored multiple senior engineers on system design, performance, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010